

▲ Practical 4 (Student)

Aim: Write a python program to demonstrate various data visualization techniques.

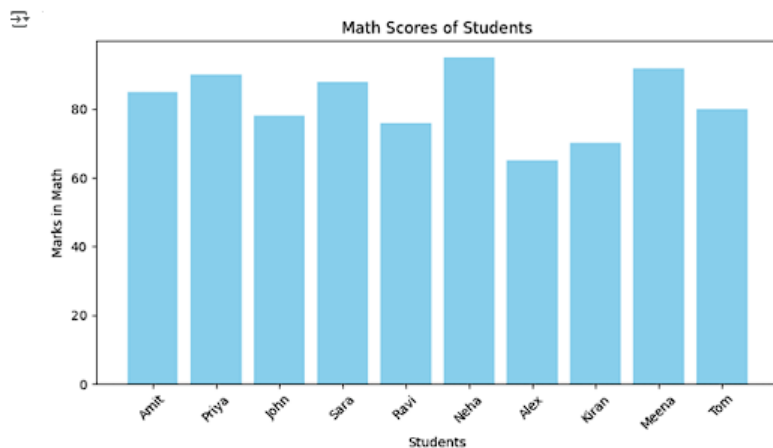
```
# Import libraries import pandas as pd import
matplotlib.pyplot as plt import seaborn as sns

# Create a custom dataset data = {
    'Name': ['Amit', 'Priya', 'John', 'Sara', 'Ravi', 'Neha', 'Alex', 'Kiran',
    'Meena', 'Tom'],
    'Math': [85, 90, 78, 88, 76, 95, 65, 70, 92, 80],
    'Science': [80, 85, 75, 89, 72, 91, 67, 78, 90, 85],
    'English': [78, 82, 74, 85, 69, 88, 72, 76, 91, 79],
    'Gender': ['M', 'F', 'M', 'F', 'M', 'F', 'M', 'F', 'M', 'M']
}

# Convert to DataFrame df = pd.DataFrame(data)
```

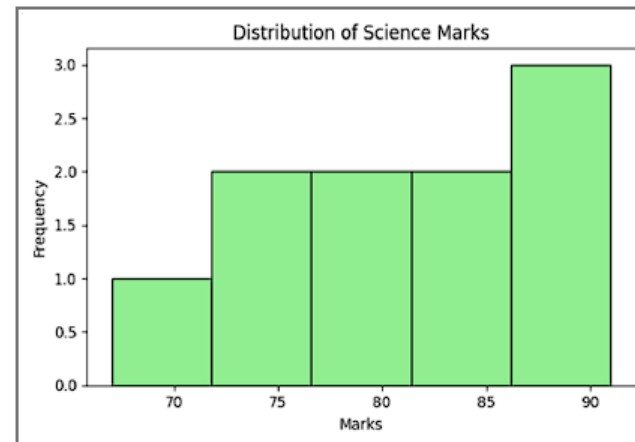
1. Bar Chart - Math Score

```
plt.figure(figsize=(8, 5)) plt.bar(df['Name'], df['Math'],
color='skyblue') plt.title('Math Scores of Students')
plt.xlabel('Students') plt.ylabel('Marks in Math')
plt.xticks(rotation=45) plt.tight_layout() plt.show()
```



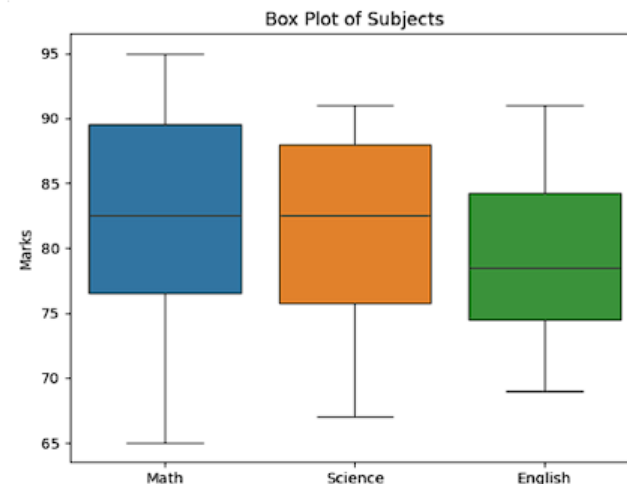
▲ 2. Histogram - Science scores

```
plt.figure(figsize=(6, 4)) plt.hist(df['Science'], bins=5,
color='lightgreen', edgecolor='black')
plt.title('Distribution of Science Marks')
plt.xlabel('Marks') plt.ylabel('Frequency')
plt.tight_layout() plt.show()
```



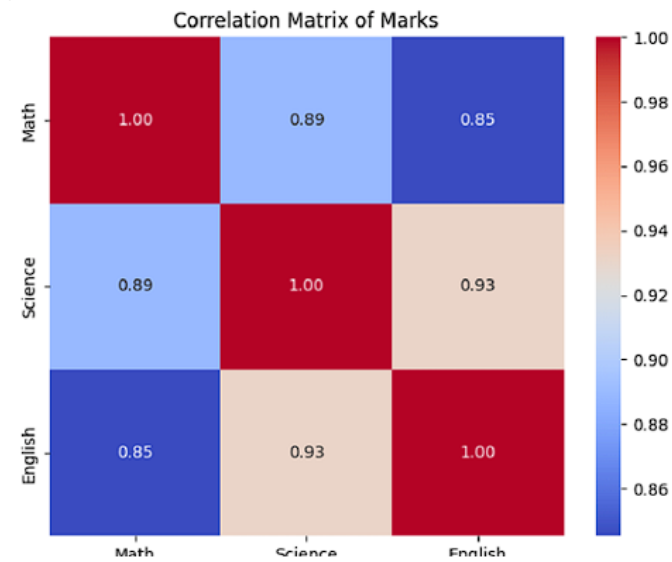
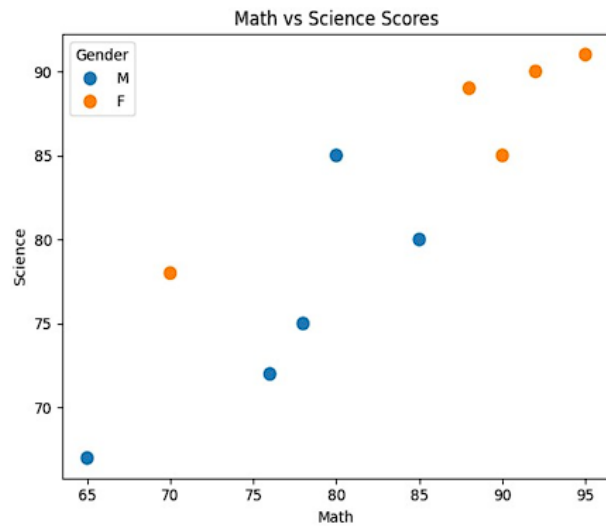
▲ 3. Box Plot - All Subjects

```
plt.figure(figsize=(6, 5)) sns.boxplot(data=df[['Math',
'Science', 'English']]) plt.title('Box Plot of Subjects')
plt.ylabel('Marks') plt.tight_layout()
plt.show()
```



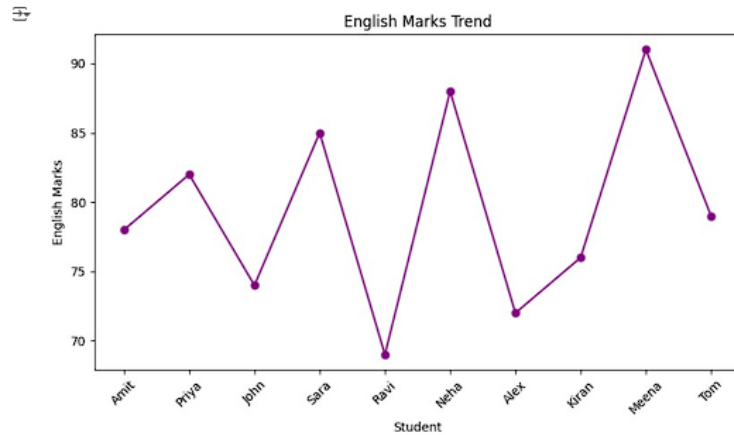
▲ 4. Scatter Plot - Math vs Science

```
plt.figure(figsize=(6, 5)) sns.scatterplot(x='Math',
y='Science', hue='Gender', data=df, s=100)
plt.title('Math vs Science Scores') plt.xlabel('Math')
plt.ylabel('Science') plt.tight_layout() plt.show()
```



5. Line Plot - English Marks Trend

```
plt.figure(figsize=(8, 5)) plt.plot(df['Name'], df['English'],
marker='o', color='purple') plt.title('English Marks Trend')
plt.xlabel('Student') plt.ylabel('English Marks')
plt.xticks(rotation=45) plt.tight_layout()
plt.show()
```



6. Heatmap - Correlation between subjects

```
plt.figure(figsize=(6, 5)) corr = df[['Math', 'Science',
'English']].corr() sns.heatmap(corr, annot=True,
cmap='coolwarm', fmt='.2f') plt.title('Correlation Matrix of
Marks') plt.tight_layout() plt.show()
```

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